

MINGXUAN LI

Causal Artificial Intelligence Lab, Columbia University, New York, NY 10027

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EDUCATION

Columbia University

Sept. 2021 - Now

Ph.D. Candidate in Computer

Graduate Research Assistant at Columbia University (09/2021-Present)

Advisor: Elias Bareinboim.

Brown University

Sept. 2019 - June 2021

Sc.M. in Computer Science, GPA: 4.0/4.0

Advisor: Michael L. Littman.

Beihang University

Sept. 2015 - June 2019

B.Sc. in Computer Science and Technology, GPA: 3.7/4.0

Advisor: Jingyuan Wang

RESEARCH INTERESTS

Causal Inference, Reinforcement Learning, Interpretability, Causal Alignment

SELECTED PUBLICATIONS

- “Confounding Robust Deep Reinforcement Learning: A Causal Approach”
NeurIPS-25 - **Mingxuan Li***, Junzhe Zhang*, Elias Bareinboim
- “Automatic Reward Sharing from Confounded Offline Data”
ICML-25 - **Mingxuan Li**, Junzhe Zhang, Elias Bareinboim
- “Causally Aligned Curriculum Learning”
ICLR-24 - **Mingxuan Li**, Junzhe Zhang, Elias Bareinboim
- “Learning Generalizable Behavior via Visual Rewrite Rules”
AAAI-22 Workshop on Reinforcement Learning in Games - Yiheng Xie*, **Mingxuan Li***, Shangqun Yu*, Michael L. Littman
- “Towards Sample Efficient Agents through Algorithmic Alignment”
AAAI-21 (Student Abstract) - **Mingxuan Li**, Michael L. Littman
- “Interpretability is a Kind of Safety: An Interpreter-based Ensemble for Adversary Defense”
KDD-20 - Jingyuan Wang, Yufan Wu, **Mingxuan Li**, Xin Lin, Junjie Wu, Chao Li

INDUSTRIAL EXPERIENCE

Uber

Jun. 2025 - Aug. 2025

Advisor: Jeffery Shen

Research Scientist Intern, Uber San Francisco

- Conducted large-scale user tipping behavior analysis and constructed a sequential user behavior dataset on billions of logs with Spark;
- Prototyped and helped productionize Epidemic Neural Recommenders (ENR) to enhance the non-linear representation learning capability of Uber’s contextual bandit platform for usage across the company;
- Prototyped and helped productionize large-scale customized Decision Transformers (DT) to model sequential user behaviors and to automate time series feature engineering; Beat the current SOTA user behavior prediction model on internal datasets by over 50%.

Amazon

Advisor: Dr. Prag Mishra

Jun. 2021 - Aug. 2021

Applied Scientist Intern, Amazon Seattle

- Analyzed billions of delivery trajectories to identify the bottleneck in delivery route planning;
- Proposed an online reinforcement learning agent that tunes the delivery route planning algorithm's hyperparameters automatically based on recent performance;
- Improved delivery efficiency by 15% compared to the previous algorithm.

Turing Microbe Co.,Ltd

Advisor: Prof. Wei Xu (IIIS, Tsinghua U)

Mar. 2019 - Jul. 2019

Computer Vision Research Intern, R&D Department

- Analyzed over 30,000 cases of gynecological diseases data with T-SNE and deep clustering to give doctors insights on new taxonomy for Bacterial Vaginal(BV) diagnosis;
- Used StyleGAN to generate realistic and highly diverse BV pictures for training young doctors;
- Highly recognized by Prof. Qinqing Liao, the chairman of the Chinese Medical Doctor Association, the gynecology branch, for insightful data analysis and practical application value of the work.

TEACHING

- Lecturer, CS 4775W, Causal Inference II, Columbia University, Spring/2025.
- TA, CS 6998, Causal Trustworthy AI, Columbia University, Fall/2023.
- TA, DATA 2040, Deep Learning and Special Topics in Data Science, Brown University, Spring/2020.

COMMUNITY SERVICES

- Organizer, Causal Reinforcement Learning Workshop, RLC, 2025
- Reviewer, ICML, 2025.
- Reviewer, NeurIPS, 2023, 2024, 2025.
- Reviewer, ICLR, 2024, 2025, 2026.
- Reviewer, AISTATS, 2025, 2026.
- Reviewer, AAAI, 2026.
- Reviewer, Journal of Machine Learning Research (JMLR), 2025
- Reviewer, International Journal of Robotics Research (IJRR), 2025
- Reviewer, Master's in Computer Science Program Admission (Columbia University), 2023.

SKILLS

Computer Languages
Software
Language

Python, Java, C/C++, SQL, HTML
L^AT_EX, Pytorch, Jax, TensorFlow
Chinese(Native), English(Fluent)

AWARDS&HONOURS

10/2018, Scholarship for Academic Achievements, Second Prize (Top 10%)
09/2018, Was selected to appear on the Dean's List for the School of Engineering, HKUST
09/2017, The 1st National Student Computer System Capability Challenge, Second Prize (Final 2/70)
05/2017, The 27th "FengRu Cup" University Students Extra-Curricular Scientific and Technological Invention Competition, Second Prize (Final 4/176)